Data Visualisation for Economics

Analysis of UK General Election 2019 Results

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Abstract

This report analyzes the 2019 UK General Election data to understand voting patterns, party performance across constituencies, and the relationship between major political parties. Key findings include a negative correlation between Labour and Conservative vote shares and regional variations in party support.

Contents

Installation of MikTex

```
options(tinytex.verbose = TRUE)
```

Installation of libraries

```
library(ukelection2019)
library(tidyverse)
library(dplyr)
library(ggplot2)
library(here)
```

Question 1 :installation of ukelection2019 package

```
remotes::install_github("kjhealy/ukelection2019")
```

```
## Skipping install of 'ukelection2019' from a github remote, the SHA1 (177cdf40) has not chang
##   Use `force = TRUE` to force installation
```

Question 2 : brief look on the table

```
ukvote2019
```

```
## # A tibble: 3,320 x 13
##    cid     constituency electorate party_name candidate votes vote_share_percent
##    <chr>   <chr>             <int> <chr>      <chr>     <int>              <dbl>
##  1 W07000~ Aberavon          50747 Labour     Stephen ~ 17008               53.8
##  2 W07000~ Aberavon          50747 Conservat~ Charlott~  6518               20.6
##  3 W07000~ Aberavon          50747 The Brexi~ Glenda D~  3108                9.8
##  4 W07000~ Aberavon          50747 Plaid Cym~ Nigel Hu~  2711                8.6
##  5 W07000~ Aberavon          50747 Liberal D~ Sheila K~  1072                3.4
##  6 W07000~ Aberavon          50747 Independe~ Captain ~   731                2.3
##  7 W07000~ Aberavon          50747 Green      Giorgia ~   450                1.4
##  8 W07000~ Aberconwy         44699 Conservat~ Robin Mi~ 14687               46.1
##  9 W07000~ Aberconwy         44699 Labour     Emily Ow~ 12653               39.7
## 10 W07000~ Aberconwy         44699 Plaid Cym~ Lisa Goo~  2704                8.5
## # i 3,310 more rows
## # i 6 more variables: vote_share_change <dbl>, total_votes_cast <int>,
## #   vrank <int>, turnout <dbl>, fname <chr>, lname <chr>
```

Question 3 : Dover constituency results

In the Dover constituency, Conservative Party won the election by 56.9% of the votes share follow by Labour party at 32.6% of the whole constiuency.

```r
dover_results <- ukvote2019 %>%
  filter(constituency == "Dover") %>%
  print()
```

```
## # A tibble: 6 x 13
##   cid      constituency electorate party_name candidate votes vote_share_percent
##   <chr>    <chr>             <int> <chr>      <chr>     <int>              <dbl>
## 1 E140006~ Dover             76355 Conservat~ Natalie ~ 28830               56.9
## 2 E140006~ Dover             76355 Labour     Charlott~ 16552               32.6
## 3 E140006~ Dover             76355 Liberal D~ Simon Do~  2895                5.7
## 4 E140006~ Dover             76355 Green      Beccy Sa~  1371                2.7
## 5 E140006~ Dover             76355 Independe~ Nathan S~   916                1.8
## 6 E140006~ Dover             76355 Women's E~ Eljai Mo~   137                0.3
## # i 6 more variables: vote_share_change <dbl>, total_votes_cast <int>,
## #   vrank <int>, turnout <dbl>, fname <chr>, lname <chr>
```

Question 4 : number of constituencies

There are 650 constituencies in UK.

```r
number_of_constituencies_df <- ukvote2019 %>%
  count(constituency)%>%
  nrow()

number_of_constituencies_df
```

```
## [1] 650
```

Question 5 : candidates of partys run in the election

All the results are not arrange. They are listed in the alphabetical order

```
candidates_per_party_alphabetical_df <- ukvote2019%>%
 group_by(party_name)%>%
 summarise(n_candidates=n())%>%
 print()
```

```
## # A tibble: 69 x 2
##    party_name                          n_candidates
##    <chr>                                      <int>
##  1 Advance Together                               5
##  2 Alliance Party                                18
##  3 Alliance for Green Socialism                   3
##  4 Animal Welfare Party                           6
##  5 Aontú                                          7
##  6 Ashfield Independents                          1
##  7 Best for Luton                                 1
##  8 Birkenhead Social Justice Party                1
##  9 British National Party                         1
## 10 Burnley & Padiham Independent Party            1
## # i 59 more rows
```

Question 6 : number of candidates for each party

The 3 main partys who run candidates are Consertive, Labour and Liberal Democrat who invest 636, 631 and 611 respectively. They are listed in descending order

```
candidates_per_party_descending_df <- ukvote2019 %>%
  group_by(party_name) %>%
  summarise(number_of_candidates = n()) %>%
  arrange(desc(number_of_candidates))

candidates_per_party_descending_df
```

```
## # A tibble: 69 x 2
##    party_name                 number_of_candidates
##    <chr>                                     <int>
##  1 Conservative                                636
##  2 Labour                                      631
##  3 Liberal Democrat                            611
##  4 Green                                       497
```

```
##  5 The Brexit Party                              275
##  6 Independent                                   224
##  7 Scottish National Party                        59
##  8 UKIP                                           44
##  9 Plaid Cymru                                    36
## 10 Christian Peoples Alliance                     29
## # i 59 more rows
```

Question 7 : percentage by all candidates fielded in each party

The 3 main partys share 56.4% in the whole country.

```
candidates_percentage_df <- ukvote2019 %>%
  group_by(party_name) %>%
  summarise(
    number_of_candidates = n(),
    percentage_of_candidates = (n() / nrow(ukvote2019)) * 100
  ) %>%
  arrange(desc(percentage_of_candidates))

candidates_percentage_df
```

```
## # A tibble: 69 x 3
##    party_name                 number_of_candidates percentage_of_candidates
##    <chr>                                     <int>                     <dbl>
##  1 Conservative                                636                    19.2
##  2 Labour                                      631                    19.0
##  3 Liberal Democrat                            611                    18.4
##  4 Green                                       497                    15.0
##  5 The Brexit Party                            275                     8.28
##  6 Independent                                 224                     6.75
##  7 Scottish National Party                      59                     1.78
##  8 UKIP                                         44                     1.33
##  9 Plaid Cymru                                  36                     1.08
## 10 Christian Peoples Alliance                   29                     0.873
## # i 59 more rows
```

Question 8 : top 5 worst candidates by votes share

The worst candidates by votes share in this election were : William Tobin (5), Bobby Elmo Smith (8), Norma Burke (22),Yace Yogeistein (23) and Scott Dore (28).

```
worst_candidates_df <- ukvote2019 %>%
  select(candidate, votes) %>%
  arrange(votes) %>%
  head(5)

worst_candidates_df
```

```
## # A tibble: 5 x 2
##   candidate        votes
##   <chr>            <int>
## 1 William Tobin        5
## 2 Bobby Elmo Smith     8
## 3 Norma Burke         22
## 4 Yace Yogenstein     23
## 5 Scott Dore          28
```

Question 9 : largest constituencies by number of votes share

We see : Bristol West (75528), Isle of Wight (74442), Wantage (67173), Cambrigeshire South (66929) and Northamptonshire South (66908).

```r
best_constituencies_df <- ukvote2019 %>%
  group_by(constituency) %>%
  summarise(total_votes = sum(votes)) %>%
  arrange(desc(total_votes)) %>%
  head(5)

best_constituencies_df
```

```
## # A tibble: 5 x 2
##   constituency           total_votes
##   <chr>                        <int>
## 1 Bristol West                 75528
## 2 Isle of Wight                74442
## 3 Wantage                      67173
## 4 Cambridgeshire South         66929
## 5 Northamptonshire South       66908
```

Question 10 : votes cast of Conservative and Labour partys

Conservative and Labour share respectively 13993282 and 10269076 votes nationally.

```r
conservatives_and_labour_votes <- ukvote2019 %>%
  filter(party_name %in% c("Labour", "Conservative")) %>%
  group_by(party_name) %>%
  summarise(total_votes = sum(votes))

conservatives_and_labour_votes
```

```
## # A tibble: 2 x 2
##   party_name   total_votes
##   <chr>              <int>
## 1 Conservative    13993282
## 2 Labour          10269076
```

Question 11 : top 10 largest vote-winning candidates measured by vote share in their constituency

> Dan Carden, in Liverpool Walton constituency is the highest vote share by percent in the country and won by 84.7%

```
top_candidates_df <- ukvote2019 %>%
  select(constituency, candidate, vote_share_percent) %>%
  arrange(desc(vote_share_percent)) %>%
  head(10)

top_candidates_df
```

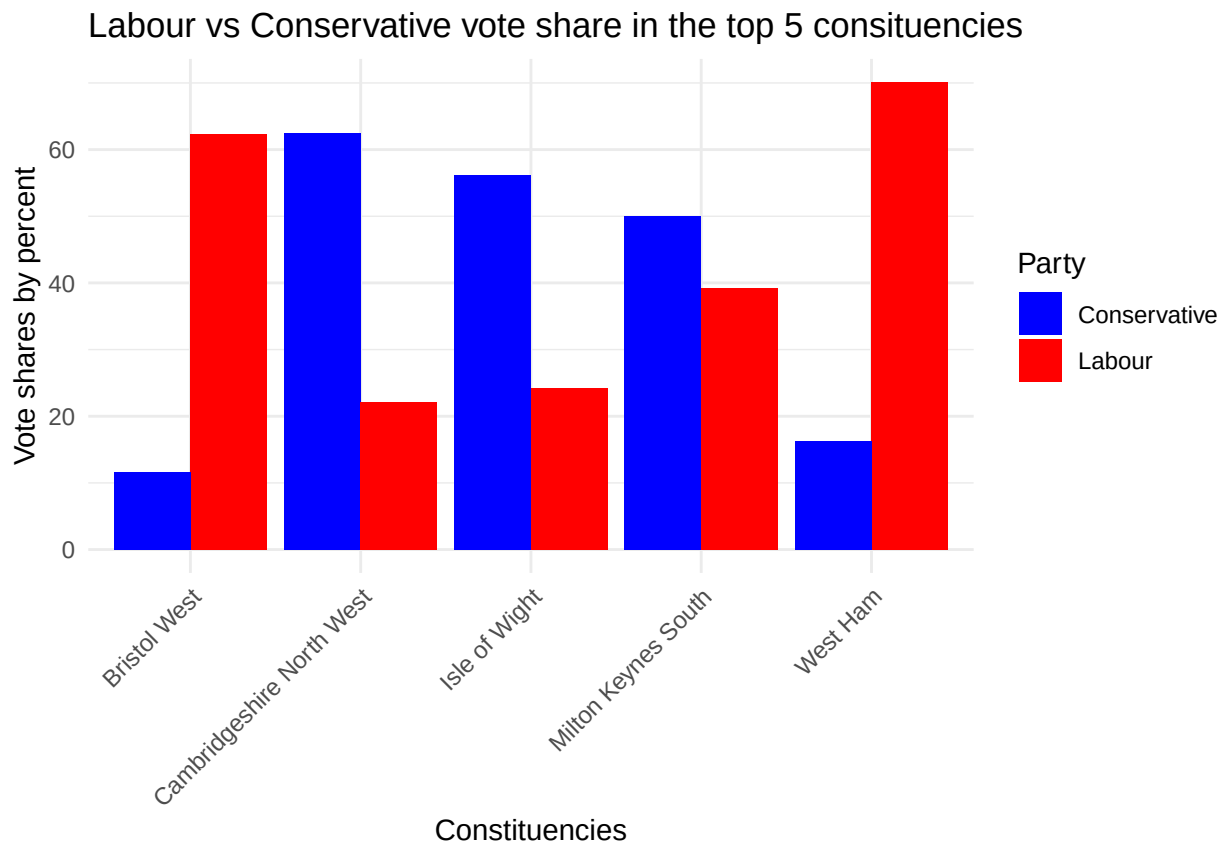
```
## # A tibble: 10 x 3
##    constituency          candidate       vote_share_percent
##    <chr>                 <chr>                        <dbl>
##  1 Liverpool Walton      Dan Carden                    84.7
##  2 Knowsley              George Howarth                80.8
##  3 Bootle                Peter Dowd                    79.4
##  4 Birmingham Ladywood   Shabana Mahmood               79.2
##  5 Birmingham Hodge Hill Liam Byrne                    78.7
##  6 Liverpool Riverside   Kim Johnson                   78
##  7 Liverpool West Derby  Ian Byrne                     77.6
##  8 Manchester Gorton     Afzal Khan                    77.6
##  9 Boston & Skegness     Matt Warman                   76.7
## 10 Castle Point          Rebecca Harris                76.7
```

Question 12 : bar graph of Conservative and Labour party in terms of electors

> Ok, Conservative party got more votes than Labour but in Bristol West and Northamptonshire South Labour won largely. Graph is save with ggsave

```
largest_constituencies_by_electors <- ukvote2019 %>%
  group_by(constituency) %>%
  summarise(total_electors = first(electorate)) %>%
  arrange(desc(total_electors)) %>%
  head(5)

top_5_constituencies <- largest_constituencies_by_electors$constituency

data_for_plot <- ukvote2019 %>%
  filter(constituency %in% top_5_constituencies) %>%
  filter(party_name %in% c("Labour", "Conservative")) %>%
  select(constituency, party_name, vote_share_percent)

q12_plot <- ggplot(data_for_plot,
                   aes(x = constituency, y = vote_share_percent, fill = party_name)) +
```

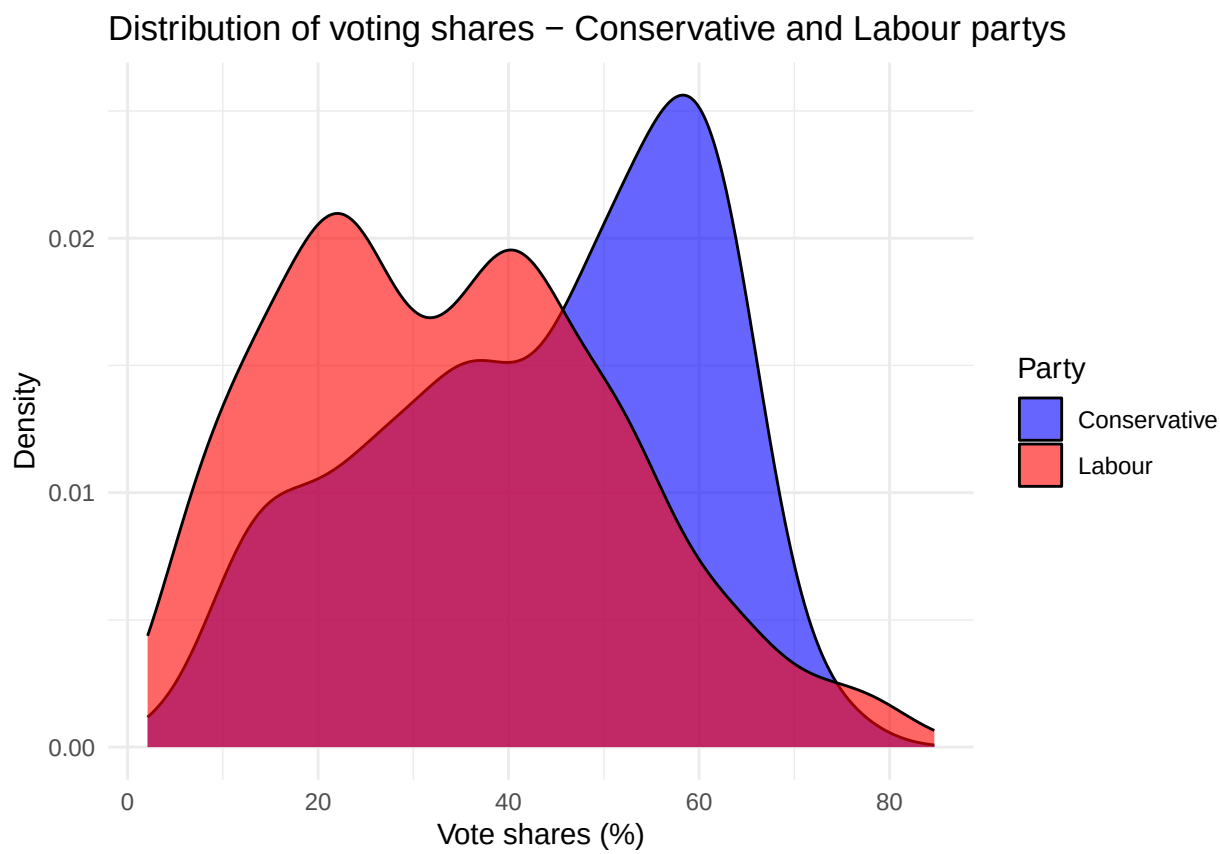
```r
  geom_col(position = "dodge") +
  labs(
    title = "Labour vs Conservative vote share in the top 5 consituencies",
    x = "Constituencies",
    y = "Vote shares by percent",
    fill = "Party"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  scale_fill_manual(values = c("Conservative" = "blue", "Labour" = "red"))

ggsave(here("graphs", "q12_labour_conservative_largest_constituencies.png"),
       plot = q12_plot, width = 10, height = 6, dpi = 300)

print(q12_plot)
```



Question 13 : distribution of votes share (Conservative and Labour)

This graph suggest that Conservative has highest values in terms of votes share by percent so much so that Labour has lowest. Same as the precedent question to see the graph.

```r
conservatives_and_labour_votes_2 <- ukvote2019 %>%
  filter(party_name %in% c("Labour", "Conservative")) %>%
  select(party_name, vote_share_percent)

q13_plot <- ggplot(conservatives_and_labour_votes_2,
                   aes(x = vote_share_percent, fill = party_name)) +
  geom_density(alpha = 0.6) +
  labs(
    title = "Distribution of voting shares - Conservative and Labour partys",
    x = "Vote shares (%)",
    y = "Density",
    fill = "Party"
  ) +
  theme_minimal() +
  scale_fill_manual(values = c("Conservative" = "blue", "Labour" = "red"))

ggsave(here("graphs", "q13_vote_share_distribution.png"),
       plot = q13_plot, width = 8, height = 6, dpi = 300)

print(q13_plot)
```



Question 14 : differences in the turnout correlation

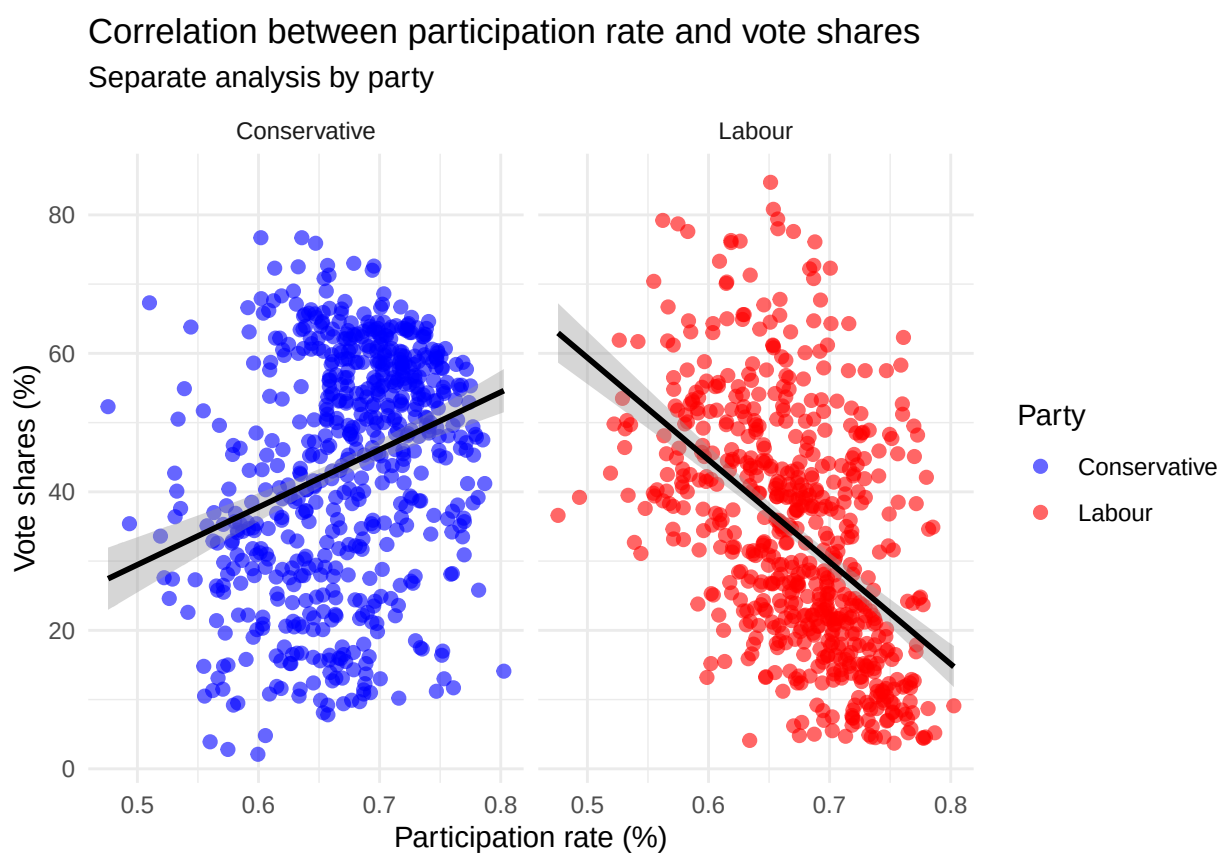
Convetive has a postive turnout correlation (0.2836349) which means that as turnout increases, so does the party's share of the vote and the opposite way for Labour party (-0.4844304).

```
turnout_data <- ukvote2019 %>%
  filter(party_name %in% c("Labour", "Conservative")) %>%
  select(constituency, party_name, vote_share_percent, turnout)

q14_plot <- ggplot(turnout_data, aes(x = turnout, y = vote_share_percent)) +
  geom_point(alpha = 0.6, size = 2, aes(color = party_name)) +
  geom_smooth(method = "lm", se = TRUE, color = "black") +
  facet_wrap(~party_name) +
  labs(
    title = "Correlation between participation rate and vote shares",
    subtitle = "Separate analysis by party",
    x = "Participation rate (%)",
    y = "Vote shares (%)",
    color = "Party"
  ) +
  theme_minimal() +
  scale_color_manual(values = c("Conservative" = "blue", "Labour" = "red"))

ggsave(here("graphs", "q14_turnout_correlation.png"),
       plot = q14_plot, width = 8, height = 6, dpi = 300)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
print(q14_plot)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
# Turnout correlation

correlation_stats <- turnout_data %>%
  group_by(party_name) %>%
  summarise(
    correlation = cor(turnout, vote_share_percent, use = "complete.obs"),
    .groups = "drop"
  )

correlation_stats
```

```
## # A tibble: 2 x 2
##   party_name   correlation
##   <chr>              <dbl>
## 1 Conservative       0.284
## 2 Labour            -0.484
```

Question 15 : top 5 constituencies for the combinaison of Green and UKIP partys

> In this top 5 we see that UKIP doesn't count anymore because it represents a very small part of the combinaison and absolutely not in 3 of them.

```r
green_ukip_data <- ukvote2019 %>%
  filter(party_name %in% c("Green", "UKIP")) %>%
  select(constituency, party_name, vote_share_percent) %>%
  pivot_wider(names_from = party_name,
              values_from = vote_share_percent, values_fill = 0) %>%
  mutate(
    Green = ifelse(is.na(Green), 0, Green),
    UKIP = ifelse(is.na(UKIP), 0, UKIP),
    combined_vote_share = Green + UKIP
  ) %>%
  arrange(desc(combined_vote_share)) %>%
  head(5)

green_ukip_data
```

```
## # A tibble: 5 x 4
##   constituency           Green  UKIP combined_vote_share
##   <chr>                  <dbl> <dbl>               <dbl>
## 1 Brighton Pavilion       57.2   0.3                57.5
## 2 Bristol West            24.9   0                  24.9
## 3 Dulwich & West Norwood  16.5   0.1                16.6
## 4 Bury St Edmunds         15.7   0                  15.7
## 5 Isle of Wight           15.2   0                  15.2
```

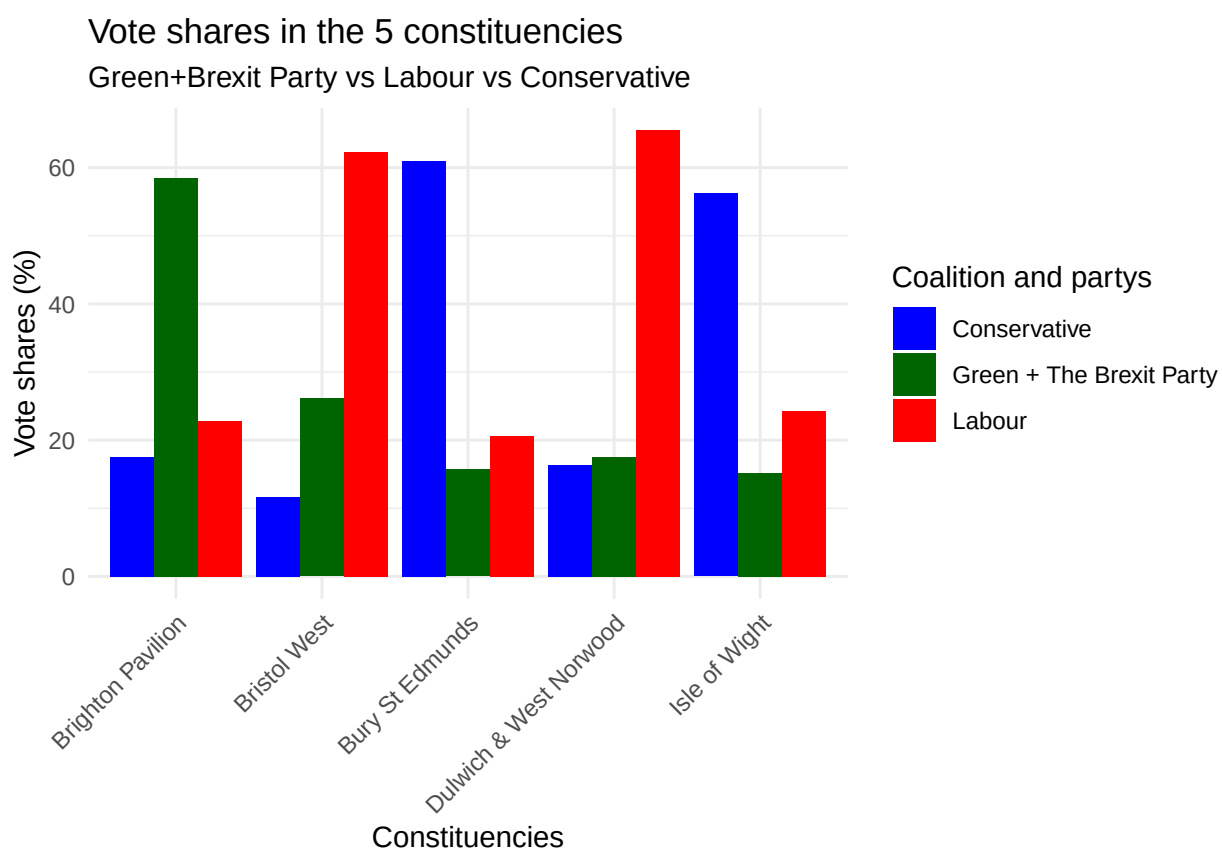
Question 16 :

i) the combined vote share of the Green and Brexit Party,
ii) the vote share of the Labour,
iii) the vote share of the Convervative.

We see that in these constituencies, the combinaison of vote shares of UKIP and Green won elections in Brighton Pavillon and arrived second in Briston West. Same way like others for the graph.

```r
top_5_constituencies <- green_ukip_data$constituency

plot_data <- ukvote2019 %>%
  filter(constituency %in% top_5_constituencies) %>%
  filter(party_name %in% c("Green", "The Brexit Party", "Labour", "Conservative")) %>%
  select(constituency, party_name, vote_share_percent)

combined_green_brexit <- plot_data %>%
  filter(party_name %in% c("Green", "The Brexit Party")) %>%
  group_by(constituency) %>%
  summarise(vote_share_percent = sum(vote_share_percent),
            .groups = "drop") %>%
```

```r
  mutate(party_name = "Green + The Brexit Party")

labour_conservative <- plot_data %>%
  filter(party_name %in% c("Labour", "Conservative"))

final_plot_data <- bind_rows(combined_green_brexit, labour_conservative)

q16_plot <- ggplot(final_plot_data,
                   aes(x = constituency, y = vote_share_percent, fill = party_name)) +
  geom_col(position = "dodge") +
  labs(
    title = "Vote shares in the 5 constituencies",
    subtitle = "Green+Brexit Party vs Labour vs Conservative",
    x = "Constituencies",
    y = "Vote shares (%)",
    fill = "Coalition and partys"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  scale_fill_manual(values = c(
    "Green + The Brexit Party" = "darkgreen",
    "Labour" = "red",
    "Conservative" = "blue"
  ))

ggsave(here("graphs", "q16_green_brexit_constituencies.png"),
       plot = q16_plot, width = 10, height = 6, dpi = 300)

print(q16_plot)
```



Question 17 : Green and Ukip less than 2%

They are 86 constituencies less than 2%.

```
green_ukip_less_2 <- ukvote2019 %>%
  filter(party_name %in% c("Green", "UKIP"))%>%
  group_by(constituency) %>%
  summarise(combined_votes_percent = sum(vote_share_percent), .groups = "drop") %>%
  filter(combined_votes_percent < 2)

nrow(green_ukip_less_2)
```

```
## [1] 86
```

Question 18 : correlation between Labour and Conservative partys by vote share

The partys are negatively correlated (-0.5752), it means that if Conservative vote share increases, the Labour tends to decrease and vice versa.

```
labour_conservative_comparison <- ukvote2019 %>%
  filter(party_name %in% c("Labour", "Conservative")) %>%
```

```r
  select(constituency, party_name, vote_share_percent) %>%
  pivot_wider(names_from = party_name,
              values_from = vote_share_percent, values_fill = 0) %>%
  mutate(
    Labour = ifelse(is.na(Labour), 0, Labour),
    Conservative = ifelse(is.na(Conservative), 0, Conservative)
  )

correlation_coefficient <- cor(labour_conservative_comparison$Conservative,
                               labour_conservative_comparison$Labour,
                               use = "complete.obs")

cat("Correlation Coefficient Labour vs Conservative:",
    round(correlation_coefficient, 4), "\n")
```

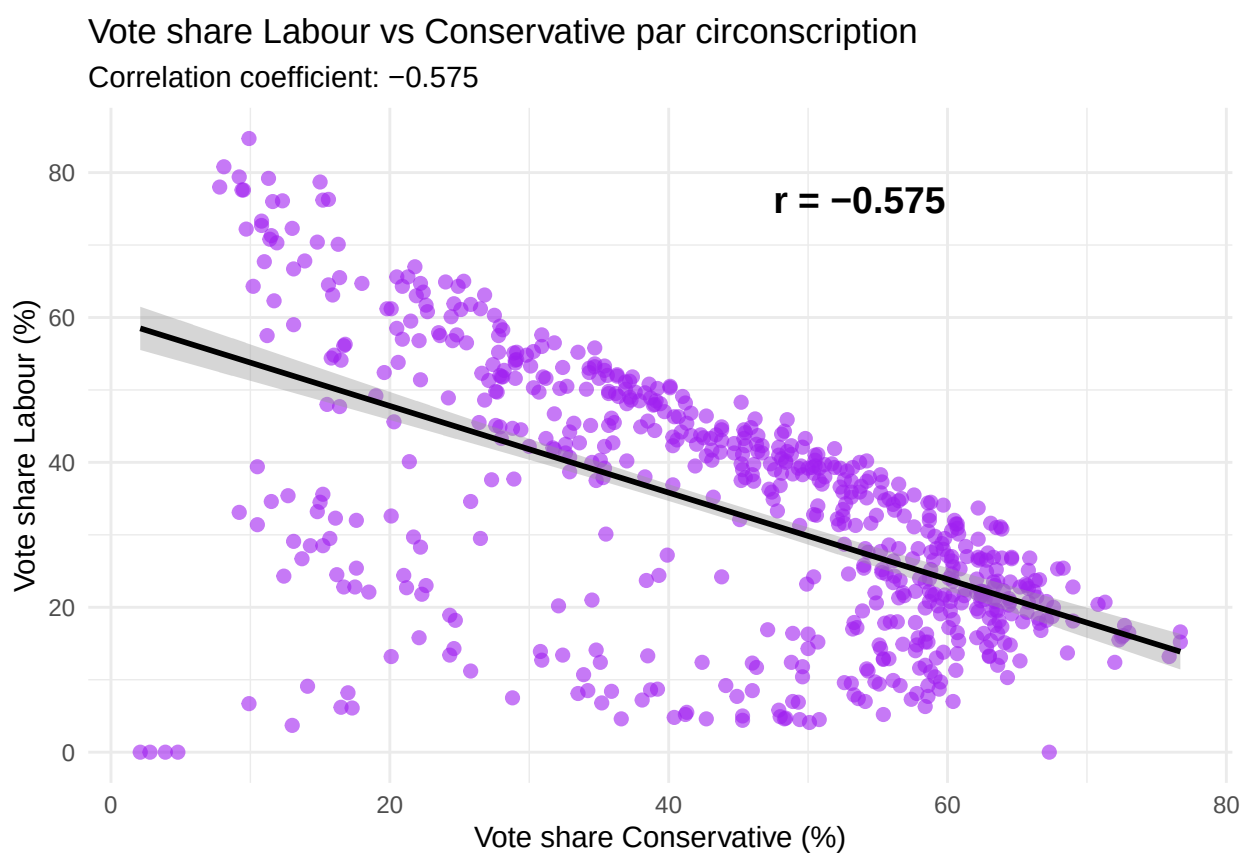
```
## Correlation Coefficient Labour vs Conservative: -0.5753
```

```r
summary_stats <- labour_conservative_comparison %>%
  summarise(correlation = cor(Conservative, Labour)
  )

q18_plot <- ggplot(labour_conservative_comparison,
                   aes(x = Conservative, y = Labour)) +
  geom_point(alpha = 0.6, size = 2, color = "purple") +
  geom_smooth(method = "lm", se = TRUE, color = "black") +
  labs(
    title = "Vote share Labour vs Conservative par circonscription",
    subtitle = paste("Correlation coefficient:", round(correlation_coefficient, 3)),
    x = "Vote share Conservative (%)",
    y = "Vote share Labour (%)"
  ) +
  theme_minimal() +
  annotate("text", x = max(labour_conservative_comparison$Conservative) * 0.7,
           y = max(labour_conservative_comparison$Labour) * 0.9,
           label = paste("r =", round(correlation_coefficient, 3)),
           size = 5, fontface = "bold")

ggsave(here("graphs", "q18_labour_conservative_correlation.png"),
       plot = q18_plot, width = 8, height = 6, dpi = 300)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```r
print(q18_plot)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Question 19 : create new dataframe

> Liberal Democrats compete directly with Labour for center-left voters with a negative correlation while showing weak alignment with Conservatives (positive correlation). This suggests they primarily split the progressive vote rather than attracting right-leaning voters.

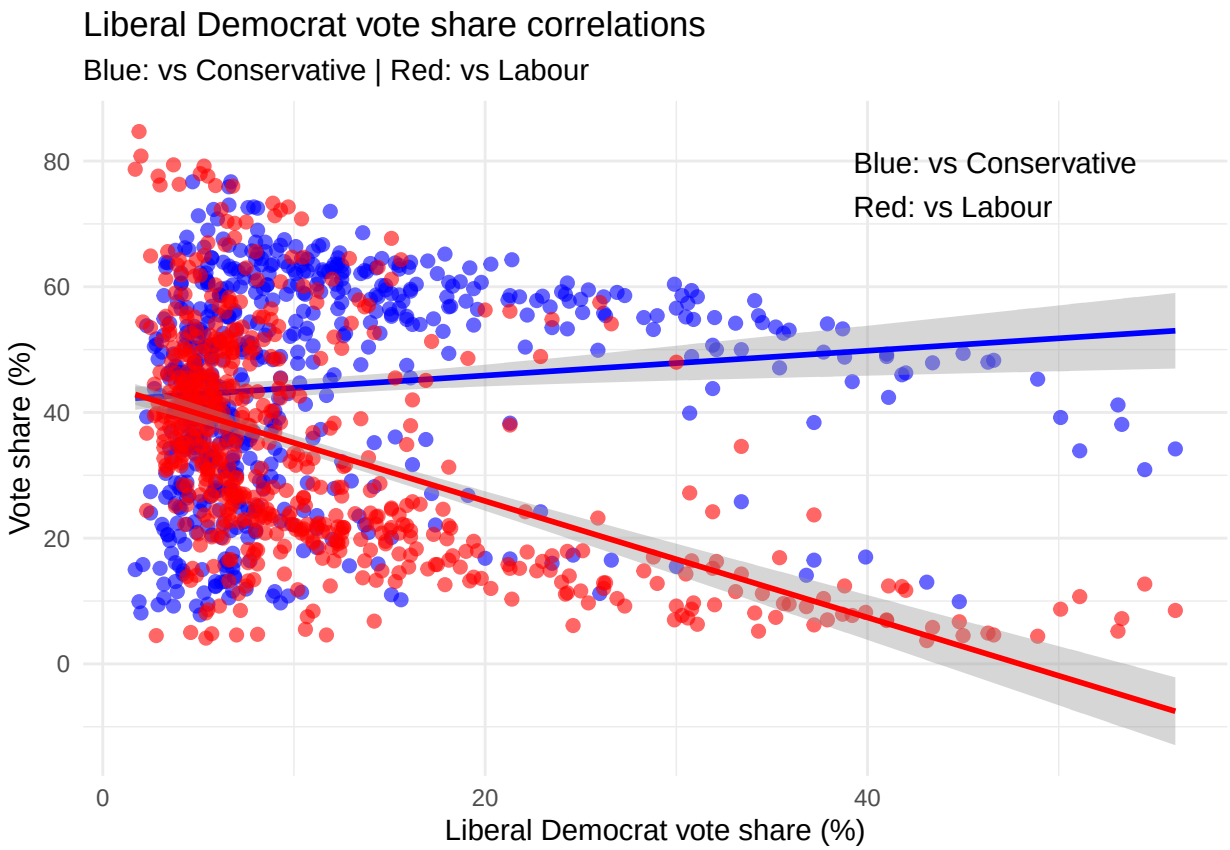
```
vote_data_wide <- ukvote2019 %>%
  filter(party_name %in% c("Liberal Democrat", "Labour", "Conservative")) %>%
  select(constituency, party_name, vote_share_percent) %>%
  pivot_wider(names_from = party_name,
              values_from = vote_share_percent, values_fill = 0) %>%
  rename_with(~gsub(" ", "_", .x))

filtered_data <- vote_data_wide %>%
  filter(Liberal_Democrat > 0)

lib_dem_conservative <- filtered_data %>%
  select(Liberal_Democrat, Conservative) %>%
  mutate(comparison = "Liberal Democrat vs Conservative")

lib_dem_labour <- filtered_data %>%
  select(Liberal_Democrat, Labour) %>%
```

```r
  rename(Conservative = Labour) %>%
  mutate(comparison = "Liberal Democrat vs Labour")

combined_data <- bind_rows(lib_dem_conservative, lib_dem_labour)

plot_q19 <- ggplot(filtered_data) +
  geom_point(aes(x = Liberal_Democrat, y = Conservative),
             color = "blue", alpha = 0.6, size = 2) +
  geom_smooth(aes(x = Liberal_Democrat, y = Conservative),
              method = "lm", se = TRUE, color = "blue") +
  geom_point(aes(x = Liberal_Democrat, y = Labour),
             color = "red", alpha = 0.6, size = 2) +
  geom_smooth(aes(x = Liberal_Democrat, y = Labour),
              method = "lm", se = TRUE, color = "red") +
  labs(
    title = "Liberal Democrat vote share correlations",
    subtitle = "Blue: vs Conservative | Red: vs Labour",
    x = "Liberal Democrat vote share (%)",
    y = "Vote share (%)"
  ) +
  theme_minimal() +
  annotate("text", x = max(filtered_data$Liberal_Democrat) * 0.7,
           y = max(c(filtered_data$Conservative, filtered_data$Labour)) * 0.9,
           label = "Blue: vs Conservative\nRed: vs Labour",
           color = "black", size = 4, hjust = 0)

ggsave(here("graphs", "q19_libdem_correlations.png"),
       plot = plot_q19, width = 8, height = 6, dpi = 300)
```

```
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
```

```r
print(plot_q19)
```

```
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
```



Question 20 : use ggsave for questions 12, 13, 14, 16, 18 et 19

0.0.1 Fin du devoir. Merci